Name: _____

TOPIC TEST 2

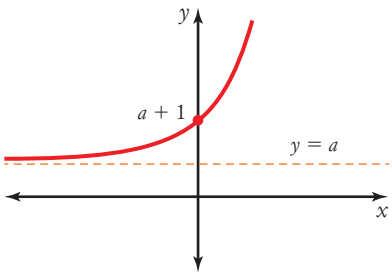
Chapter 11 Exponential functions

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 6 multiple-choice questions (6 marks)
- Part B: 7 short answer questions (27 marks)
- Part C: 4 analysis questions (21 marks)

Part A

6 multiple-choice questions
1 mark each: 6 marks
Circle the correct answer.

1 Which of the following equations best corresponds to the graph shown on the right?



A $y = 2^{-x} + a$

B $y = 2^{x} + a$

C $y = 2^{(a+1)x} + a$

D $y = (a + 1)2^{-x} + a$

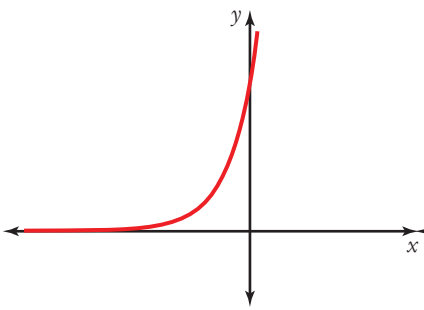
E $y = 2^{x} - a$

[1 mark]

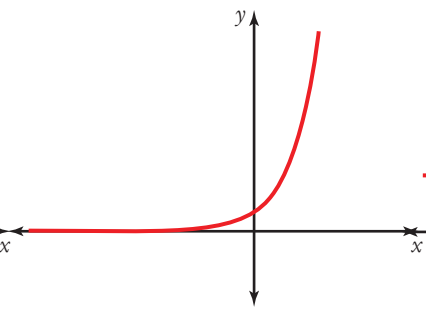
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Use the graphs below to answer questions **2**–**4**.

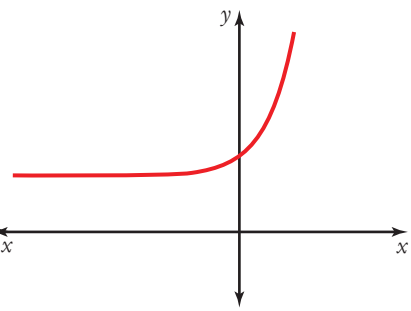
A



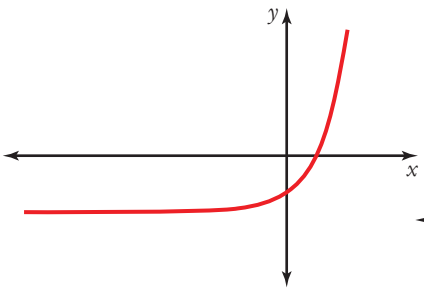
B



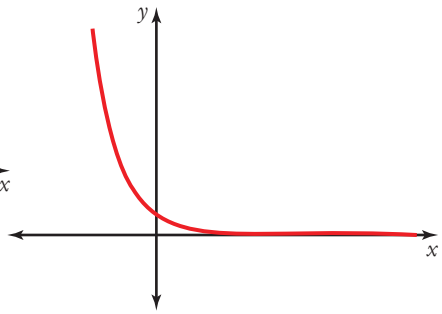
C



D



E



2 Which of the graphs **A**–**E** above could represent $y = a^x + b$, where $a > 1$ and $b = 0$?

[1 mark]

3 Which of the graphs **A**–**E** above could represent $y = a^{x+b}$, where $a > 1$ and $b > 0$?

[1 mark]

4 Which of the graphs **A**–**E** above could represent $y = a^x + b$, where $a > 1$ and $b < 0$?

[1 mark]

5 The solution of $3 \times 10^{3x+1} = 72$ is closest to:

A $x = 0.13$

B $x = 2.07$

C $x = 1.71$

D $x = 5.18$

E $x = 0.46$

[1 mark]

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6 Which of the following tables shows data from an exponential rule?

A

x	y
0	500
1	700
2	1000
3	1400

B

x	y
0	500
1	400
2	320
3	256

C

x	y
0	500
1	400
2	300
3	200

D

x	y
0	500
1	600
2	700
3	800

E

x	y
0	500
1	350
2	250
3	200

[1 mark]

Part B

7 short answer questions
27 marks
Show your working where appropriate.

1 Evaluate $\sqrt[3]{\dfrac{8.3 \times 4.1}{0.2 + 5.4 \div 1.3}}$, correct to 3 significant figures.

[2 marks]

2 Simplify and express with positive indices.

a $\left(\sqrt[4]{g^3}\right)^{\frac{1}{2}} \times \left(\sqrt[3]{g^7}\right)^{\frac{3}{4}} \times g^{-\frac{5}{8}}$

b $\dfrac{y^{\frac{1}{2}} z^{-\frac{2}{3}}}{\sqrt{16yz^6}} \times \sqrt[3]{z^{-1}}$

[6 marks]

3 $g(h) = 2.4 \times (0.8)^h - 5$. Find the value of:

a $g(4)$

b $g(-2)$

[2 marks]

4 Solve the following equations.

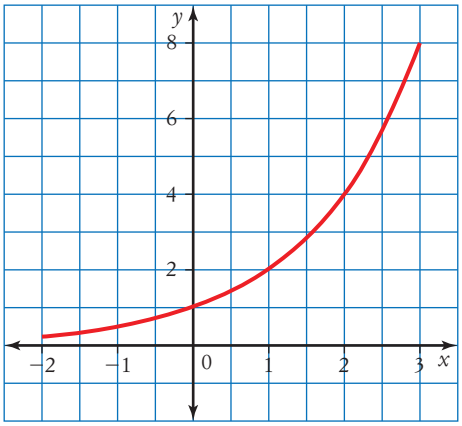
a $9^{x-5} = 243$

b $5^{3-2x} - 125 = 0$

c $49 - 343^{2x+3} = 0$

[9 marks]

5 The graph of $y = 2^x$ is shown. Use the graph to solve $2^x = 3$



[2 marks]

6 Use a graph drawn with a graphics calculator to solve $5^x = 16$.

[2 marks]

7 For each of the following exponential functions, calculate the y-intercept and the equation of the horizontal asymptote.

a $y = 3 \times 5^{(x-1)}$

b $y = 6 \times 1.4^{(x-2)} + 4$

[4 marks]

Part C

4 analysis questions
21 marks
Show your working where appropriate.

1 Sketch the graph of each of the following by first determining the key features of the graph.

a $y = 5^x - 4$

b $y = 3^{x+2} + 2$

[6 marks]

2 Sketch the graph of $y = 3^{x-4} + 3$ by transforming the graph of $y = 3^x$.

[4 marks]

3 The size of a fungus doubles every day from an initial mass of 0.04 g.

a Set up an exponential model of the mass of the fungus.

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b Find the mass of the fungus after:

i 3 days _____

ii 10 days _____

iii 25 days _____

c Find the time taken for the mass to reach 500 g.

[6 marks]

4 A sample of radioactive material loses 4% of its radioactivity per annum. Calculate the half-life of the material.

[5 marks]
Total marks: 54

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Answers

Part A **[1 mark each]**

1 The graph is the same as $y = 2^x$ but translated a units up. This would be $y = 2^x + a$.

$\therefore$ **B** **[1 mark]**

2 The graph of $y = a^x + b$ will look like the graph of $y = a^x$ (where $a > 1$) because $b = 0$.

$\therefore$ **B** **[1 mark]**

3 The graph of $y = a^{x+b}$ will look like the graph of $y = a^x$ (where $a > 1$) that has been translated b units to the left because $b > 0$.

$\therefore$ **A** **[1 mark]**

4 The graph of $y = a^x + b$ will look like the graph of $y = a^x$ (where $a > 1$) that has been translated b units down because $b < 0$.

$\therefore$ **D** **[1 mark]**

5 The equation can be simplified to $10^{3x+1} = 24$. Since 24 is between 10^1 and 10^2, $3x + 1$ must lie between 1 and 2. In other words, $1 < 3x + 1 < 2$. This can be simplified to $0 < 3x < 1$ and $0 < x < \frac{1}{3}$. The only choice that meets this criteria is **A**.

$\therefore$ **A** **[1 mark]**

6 For the rule to be exponential, the quotient of successive terms must be constant. This is only true for **B**.

$\therefore$ **B** **[1 mark]**

Part B

1 $\sqrt[3]{\dfrac{8.3 \times 4.1}{0.2 + 5.4 \div 1.3}} = \sqrt[3]{\dfrac{34.03}{4.3538...}} = 1.9845... \approx 1.98$ **[2 marks]**

2 a $\left(\sqrt[4]{g^3}\right)^{\frac{1}{2}} \times \left(\sqrt[3]{g^7}\right)^{\frac{3}{4}} \times g^{-\frac{5}{8}} = g^{\frac{3}{8}} \times g^{\frac{21}{12}} \times g^{-\frac{5}{8}} = g^{\frac{7}{4}} \times g^{-\frac{1}{4}} = g^{\frac{3}{2}}$ **[3 marks]**

b $\dfrac{y^{\frac{1}{2}} z^{-\frac{2}{3}}}{\sqrt{16yz^6}} \times \sqrt[3]{z^{-1}} = \dfrac{y^{\frac{1}{2}} z^{-\frac{2}{3}}}{4} \times z^{-\frac{1}{3}} \times y^{-\frac{1}{2}} \times z^{-3} = \dfrac{y^0 z^{-1-3}}{4} = \dfrac{1}{4z^4}$ **[3 marks]**

[6 marks]

3 a $g(4) = 2.4 \times (0.8)^4 - 5 \approx -4.0170$ **[1 mark]**

b $g(-2) = 2.4 \times (0.8)^{-2} - 5 \approx -1.25$ **[1 mark]**

[2 marks]

4 a $9^{x-5} = 243 \Rightarrow 3^{2(x-5)} = 3^5$ **[1 mark]**

$\Rightarrow 2x - 10 = 5 \Rightarrow x = 7\frac{1}{2}$ **[2 marks]**

b $5^{3-2x} - 125 = 0 \Rightarrow 5^{3-2x} - 5^3 = 0$ **[1 mark]**

$\Rightarrow 3 - 2x = 3 \Rightarrow x = 0$ **[2 marks]**

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c $49 - 343^{2x+3} = 0 \Rightarrow 7^2 - 7^{3(2x+3)} = 0$ **[1 mark]**

$\Rightarrow 2 = 6x + 9 \Rightarrow x = -1\frac{1}{6}$ **[2 marks]**

[9 marks]

5 Locate the point on the graph where $y = 3$.

Draw a vertical line to intersect the x-axis and read the value of x.

$x \approx 1.6$

[2 marks]

6 Draw the graph. Use the trace function or locate the zero of $5^x - 16 = 0$.

$x \approx 1.723$ **[2 marks]**

7 **a** Let $x = 0$: $y = 3 \times 5^{-1} = 0.6$

y-intercept $= 0.6$ **[1 mark]**

Compare with $y = Aa^{x+B} + C$: $C = 0$, so horizontal asymptote is $y = 0$ **[1 mark]**

b Let $x = 0$: $y = 6 \times 1.4^{(1-2)} + 4 = 7.06$

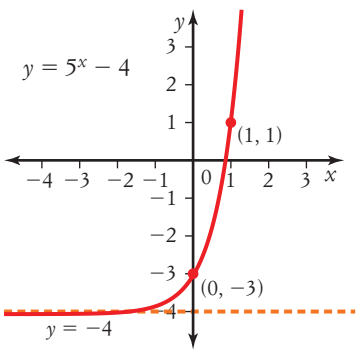
y-intercept $= 7.06$ **[1 mark]**

Compare with $y = Aa^{x+B} + C$: $C = 4$, so horizontal asymptote is $y = 4$ **[1 mark]**

[4 marks]

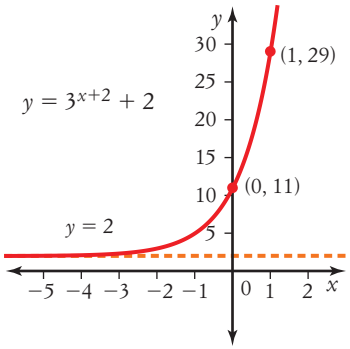
Part C

1 **a**



[3 marks]

b

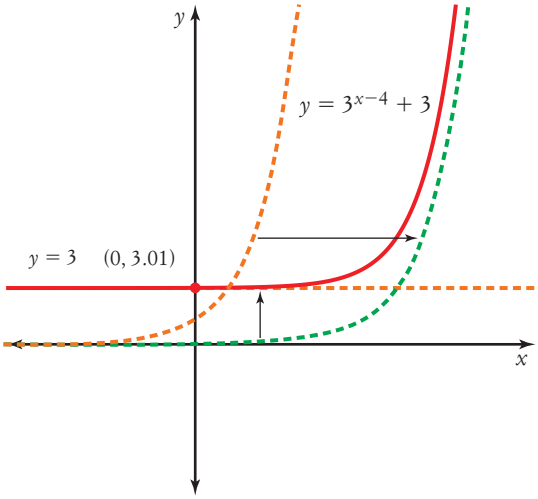


[3 marks]

[6 marks]

2 Compare $y = 3^{x-4} + 3$ with $y = a^{x+B} - C$. $B = -4$ and $C = 3$. **[1 mark]**

Plot the graph of $y = 3x$. Slide the graph 4 units to the right and then slide the graph 3 units up. **[1 mark]**



[2 marks]

[4 marks]

3 a Growth rate = 2; initial value = 0.04. Let d = number of days. **[1 mark]**

$F = 0.04 \times 2^d$ **[1 mark]**

b i $F = 0.04 \times 2^3 = 0.32$ g **[1 mark]**

ii $F = 0.04 \times 2^{10} = 40.96$ g **[1 mark]**

iii $F = 0.04 \times 2^{25} = 1\,342\,177$ g **[1 mark]**

c $500 = 0.04 \times 2^d$

$2^d = 12\,500$ **[1 mark]**

Use trial and error to solve for d, or use the **solve()** function on your CAS calculator.

$d \approx 13.6$ days **[1 mark]**

[6 marks]

4 Growth factor = $(100 - 4)\% = 0.96$; initial value = a **[1 mark]**

Let y = number of years

Let R = radioactivity

$R = a \times 0.96^y$ **[1 mark]**

At half-life, $R = \frac{1}{2}a$ **[1 mark]**

$\frac{1}{2}a = a \times 0.96^y$

$\frac{1}{2} = 0.96^y$ **[1 mark]**

Use trial and error to solve for y, or use the **solve()** function on your CAS calculator.

$y \approx 16.98$ years **[1 mark]**

[5 marks]

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